Changing velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate acceleration and distinguish speeding up, slowing down, and turning.

SUCCESS CRITERIA - I can...

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A velocity graph is constant at -4 m/s for 6 s. Find displacement. A second object has components 8 m/s east and 3 m/s south; sketch its direction.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_{avg} = (v_f - v_i)/\Delta t$.
i = initial; f = final. Compare directions of velocity and acceleration to decide whether speed increases.

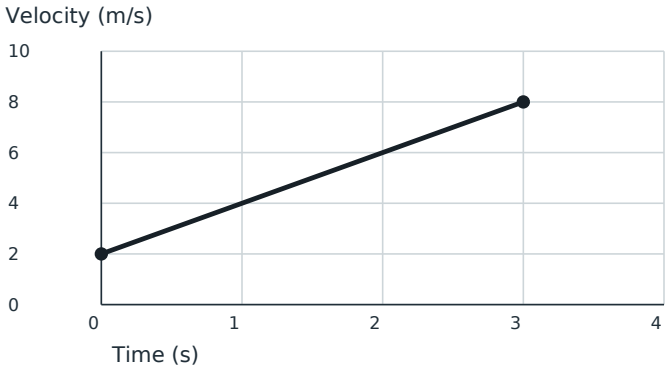
Worked reasoning: A cart moving east slows from 8 to 2 m/s in 3 s. Taking east positive, $a=(2-8)/3=-2$ m/s². It keeps moving east while acceleration points west.

Your turn: Draw initial and final velocity arrows and one acceleration arrow. Use arrow lengths to show the speed change.

Marzano check - circle: ① ② ③

03 Watch velocity grow

Read the graph. Find velocity change over 3 s, then acceleration. Is the cart speeding up?



Marzano check - circle: ① ② ③

04 A bike slows

A bike moving east slows from 10 m/s to 4 m/s in 5 s. How much speed is lost each second? Which way does acceleration point?

Support: Step 1: Find the speed lost: 10 - 4. Step 2: Divide by 5 s. Step 3: Acceleration points opposite the bike's motion while it slows.

Marzano check - circle: ① ② ③

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Continue down each column.

05 A turn at steady speed

A toy car moves around a circle at steady speed. Does it accelerate? Draw velocity arrows at two points to support your answer.

Marzano check - circle: 1 2 3

06 Choose a method

A goes 5 m east and 3 m north. B then goes 2 m west and 7 m south. Draw the route and find the endpoint relative to the start.

Marzano check - circle: 1 2 3

07 Exit check

A cart moves west at 6 m/s, then west at 2 m/s two seconds later. Is it speeding up or slowing down? Find the acceleration direction and magnitude.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Make a velocity-time graph of a cart speeding up westward from 1 to 5 m/s in 2 s. Use east as positive and explain the acceleration sign.

Marzano check - circle: 1 2 3